Lab 6 — IPv4 Subnetting and Address Planning with IP Calculator

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 02:** Networking — Core 1, 23% of Core 1
> **Exam objective:** Calculate subnet masks, network and broadcast addresses, usable host ranges and CIDR notation, and design an address plan for a SOHO network (Core 1 objective 2.5).

Goal

You use IP Calculator to work subnetting from both directions — decoding a given address and designing a plan to meet a host requirement — then verify every result by hand so you can reproduce it in an exam where no calculator is allowed.

What you'll produce

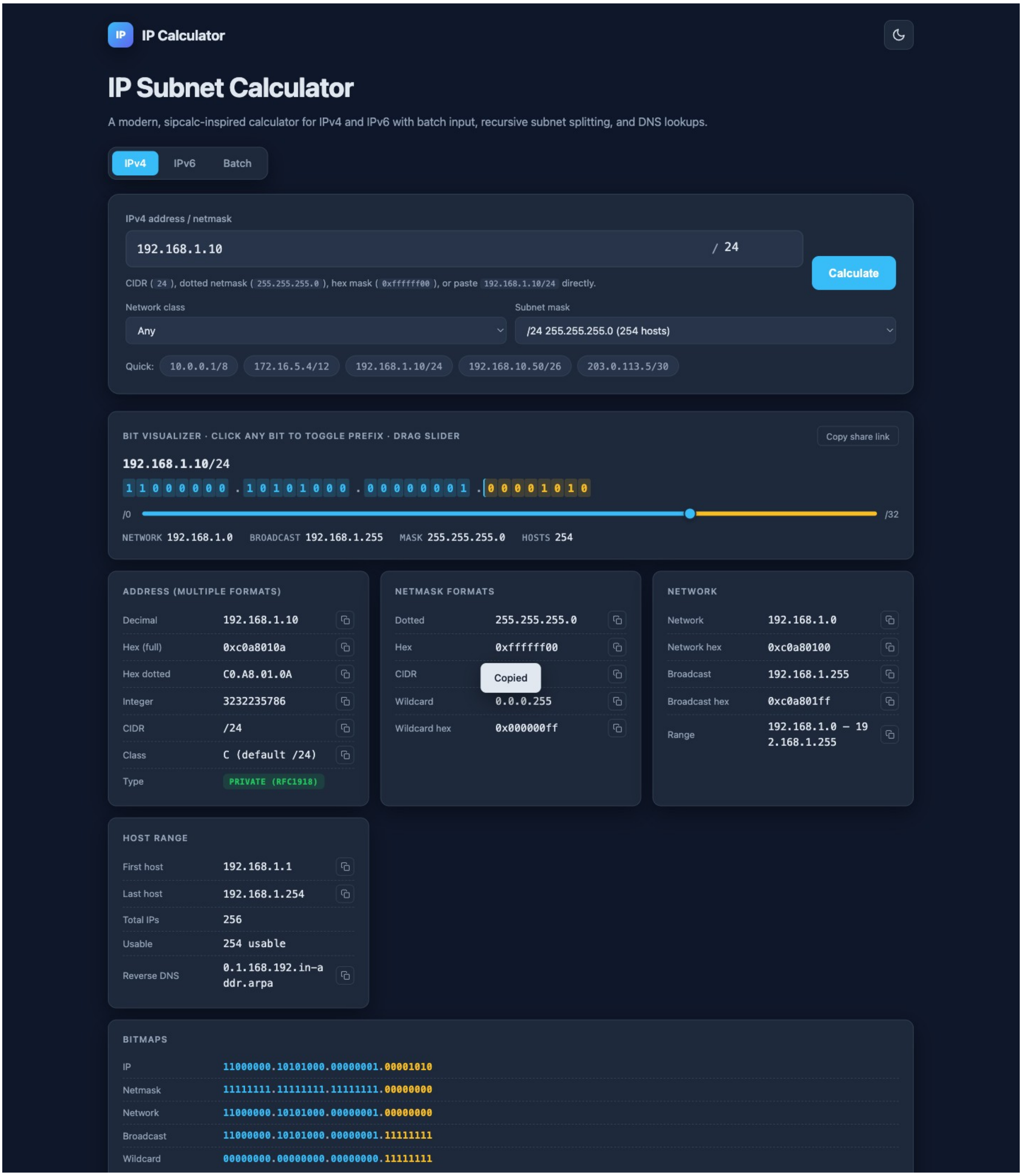
A verified subnetting worksheet and a complete four-subnet SOHO address plan exported as CSV.

Tools and equipment

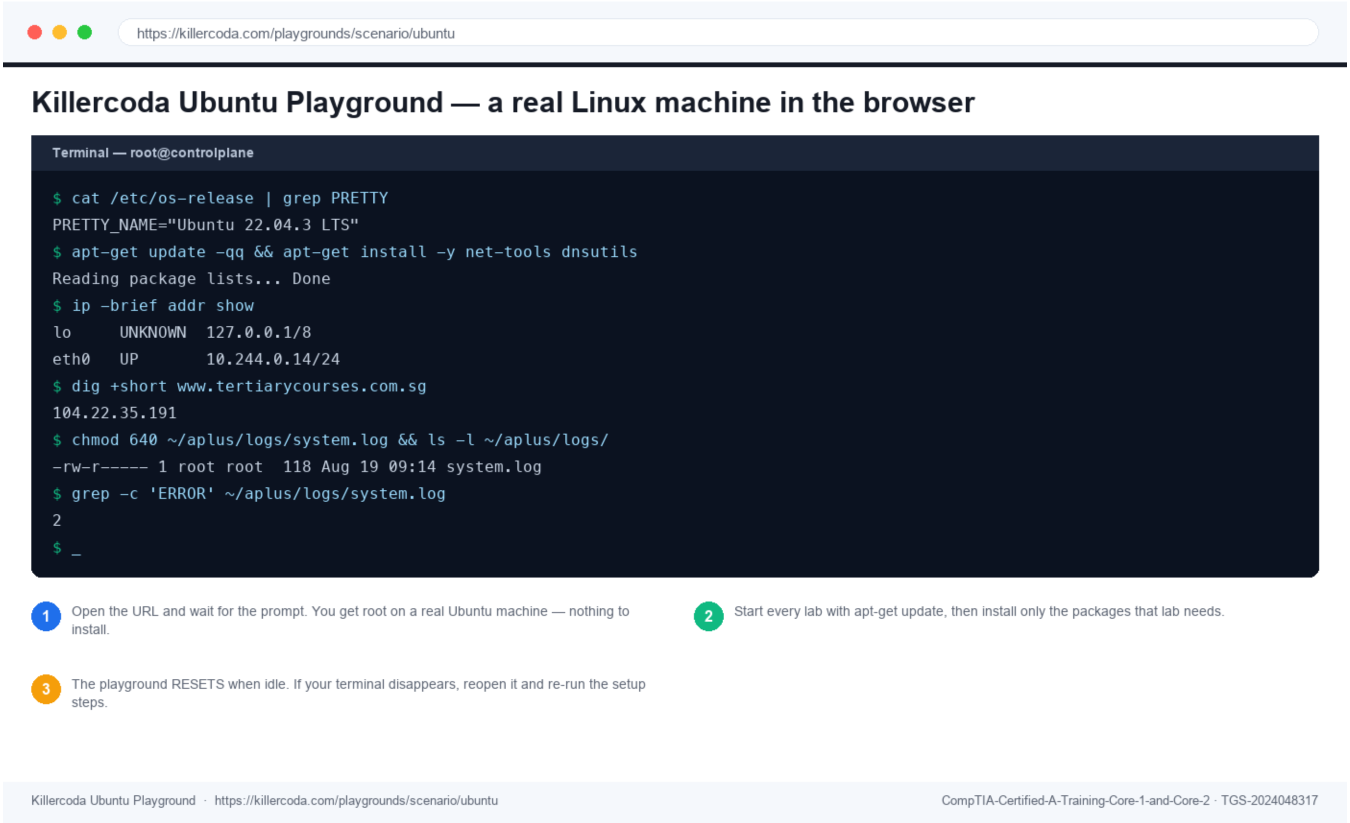
IP Calculator (https://alfredang.github.io/ipcalculator/), Killercoda Ubuntu Playground, ipcalc

Browser tools used in this lab

- **IP Calculator** — https://alfredang.github.io/ipcalculator/
- **Killercoda Ubuntu Playground** — https://killercoda.com/playgrounds/scenario/ubuntu



IP Calculator — the panels and fields this lab uses.



Killercoda Ubuntu Playground — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open IP Calculator at https://alfredang.github.io/ipcalculator/ and select the IPv4 tab.
2. Enter 192.168.10.75/24 and record the network address, broadcast address, first and last usable host, netmask in dotted decimal, and the usable host count.
3. Change the prefix to /26 with the same address and record how the network address, broadcast address and usable host count change.
4. Verify the /26 result by hand: 32 minus 26 leaves 6 host bits, 2 to the power 6 is 64 addresses per block, minus 2 reserved gives 62 usable hosts. Confirm this matches the tool.
5. Enter 172.16.0.0/12 and 10.0.0.0/8 in turn, and confirm from the output that all three RFC 1918 private ranges are non-routable on the internet.

6. Enter 169.254.14.9/16 and record what this range signifies — an APIPA address assigned when no DHCP server answered.
7. Design a SOHO plan from the 192.168.20.0/24 block for four departments needing 50, 25, 10 and 5 hosts. Choose the smallest prefix that satisfies each requirement.
8. Verify each chosen prefix in IP Calculator: /26 gives 62 usable for the 50-host subnet, /27 gives 30 for the 25-host subnet, /28 gives 14 for the 10-host subnet and /29 gives 6 for the 5-host subnet.
9. Switch to the Batch tab, paste all four subnets one per line with a comment naming each department, and process them together.
10. Export the batch results to CSV using the export control, and keep the file as the address-plan deliverable.
11. Open the Killercoda Ubuntu playground and install ipcalc to cross-check your work with an independent tool.

```
apt-get update -qq && apt-get install -y ipcalc
```

12. Cross-check the first subnet on the command line and confirm the network, broadcast and host range match IP Calculator exactly.

```
ipcalc 192.168.20.0/26
```

Test it — verification

Every subnet in your plan satisfies its host requirement with no wasted block larger than necessary, the hand calculation matches the IP Calculator output for the /26 case, ipcalc agrees with the browser tool, and the CSV export contains all four subnets.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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